

pii-quiet-alarm — precision-first sensitive-data tagging

Run pii-quiet-alarm-2026-08-25 · commit 3ce3c8c · 1,000 trials · 24 measured arms (3 models × 8 corpora) · MLflow experiment pii-quiet-alarm

Results

Headline — equal-corpus, sealed evaluation

Arm	priority macro F0.5	macro F0.5	doc precision	doc specificity	doc recall	one-core p95	verdict
quiet-cascade	0.8233	0.7766	0.8806	0.8693	0.7729	3.93 ms	blocked
quiet-nogate (ablation)	0.8205	0.7746	0.7473	0.5783	0.8943	3.97 ms	blocked
champion_1k (prior champion)	0.2057	0.4163	0.6162	0.0005	0.9994	1.81 ms	blocked

quiet-nogate carries **identical weights and thresholds** to quiet-cascade with the document gate disabled, so every difference between those two rows is the gate alone.

The gate ladder — why nothing shipped

Each constraint is judged per corpus on the bootstrap **lower bound**, with a 30-instance minimum. n/m is passing over measurable.

Arm	doc precision ≥ 0.90	doc specificity ≥ 0.85	doc recall ≥ 0.85	priority tag recall ≥ 0.75	p95 ≤ 5 ms
quiet-cascade	FAIL 1/3 (worst 0.7893)	FAIL 2/3 (worst 0.8299)	FAIL 1/3 (worst 0.6604)	FAIL 36/55 (worst 0.0000)	PASS
quiet-nogate	FAIL 1/3 (0.5892)	FAIL 0/3 (0.4533)	FAIL 2/3 (0.8229)	FAIL 36/55 (0.0000)	PASS
champion_1k	FAIL 0/3 (0.4632)	FAIL 0/3 (0.0000)	PASS 3/3	PASS 55/55	PASS

No feasible arm. Promotion refused; no champion packaged.

Document level, per corpus — the three that hold genuine negatives

Point estimate with the 95% bootstrap lower bound in brackets. The other five corpora have prevalence 1.0 and report these as NOT_MEASURABLE, never as a pass.

Arm	Corpus	precision	specificity	recall
quiet-cascade	pii2 (synthetic)	0.9783 (0.9764)	0.8926 (0.8841)	0.9324 (0.9295)
quiet-cascade	datax (real)	0.8089 (0.7893)	0.8462 (0.8299)	0.7104 (0.6905)

Arm	Corpus	precision	specificity	recall
quiet-cascade	govdocs2 (real)	0.8545 (0.8414)	0.8692 (0.8574)	0.6759 (0.6604)
champion_1k	pii2 (synthetic)	0.8383 (0.8342)	0.0000 (0.0000)	1.0000 (1.0000)
champion_1k	datax (real)	0.4783 (0.4632)	0.0000 (0.0000)	1.0000 (1.0000)
champion_1k	govdocs2 (real)	0.5320 (0.5205)	0.0016 (0.0003)	0.9983 (0.9968)

The split is the finding: on synthetic documents the cascade clears every document bar comfortably; on **real business documents it stays shut too often** (recall 0.71 and 0.68 against a 0.85 floor).

Priority-tag quality, per corpus

Corpus	quiet-cascade	champion_1k
pii2	F0.5 0.965 · P 0.966 · R 0.967	F0.5 0.045 · P 0.038 · R 1.000
pii_holdout	F0.5 0.942 · P 0.964 · R 0.864	F0.5 0.185 · P 0.159 · R 0.999
openpii	F0.5 0.907 · P 0.940 · R 0.799	F0.5 0.385 · P 0.341 · R 1.000
ai4privacy	F0.5 0.880 · P 0.929 · R 0.742	F0.5 0.184 · P 0.157 · R 1.000
betterdataai	F0.5 0.422 · P 0.632 · R 0.319	F0.5 0.228 · P 0.218 · R 1.000

pii2 is the flattering corpus (0.965); betterdataai is the weakest (0.422). Quote the equal-corpus **0.8233**, not either end.

Prediction rate — the share of documents receiving any tag — falls from **0.9998** (champion_1k tags essentially every document) to **0.7876**.

The 19 failing priority pairs

Worst first, of 55 measurable pairs. Two clusters, with different causes.

Tag	Corpus	recall	ci_low	support
medical_record_number_mrn	betterdataai	0.0132	0.0000	151
password	betterdataai	0.1000	0.0200	50
address	betterdataai	0.3110	0.2559	254
address	govdocs2	0.3636	0.3322	891
personal_identification_number_pin	ai4privacy	0.5172	0.4345	145
address	datax	0.5600	0.5086	350
full_name	govdocs2	0.6208	0.6037	3,215
driver_s_license_number	openpii	0.6206	0.6102	8,538
full_name	ai4privacy	0.6331	0.6192	5,470

Tag	Corpus	recall	ci_low	support
password	ai4privacy	0.7020	0.6779	1,245
social_security_number	ai4privacy	0.7113	0.6925	2,400
passport_number	ai4privacy	0.7205	0.6942	1,102
address	pii2	0.7286	0.7128	2,841
driver_s_license_number	pii_holdout	0.7315	0.7147	2,808
military_identification_number	openpii	0.7307	0.7237	13,763
passport_number	openpii	0.7363	0.7284	13,763
address	ai4privacy	0.7461	0.7295	2,529
visa_number	openpii	0.7404	0.7334	13,763
driver_s_license_number	ai4privacy	0.7657	0.7393	1,101

Cluster A — marginal (11 pairs, 0.70–0.77). These sit just under a 0.75 floor. Their thresholds were chosen to hit exactly 0.75 on *pooled* held-in calibration, so any distribution shift lands them on the wrong side. This is a calibration-transfer problem, not a capability limit.

Cluster B — structural (8 pairs). address fails on all five corpora it is measurable on, and MRN/password collapse on betterdataai specifically. Address is the tag the frozen collapse merged street_number_and_name into, and betterdataai carries silver (model-generated) labels.

TL;DR

The precision problem was a **data defect, not a model defect**, and fixing it moved the numbers a long way — but the run did not clear the bars it set for itself, so nothing was promoted.

- **Priority macro F0.5: 0.2057 → 0.8233** (4.0×). Priority-tag precision on the weakest complete corpus went from 0.218 to 0.632; on the strongest, 0.038 to 0.966.
 - **Document specificity: 0.0005 → 0.8693.** The prior champion flagged 99.95% of genuinely PII-free documents; the cascade correctly stays silent on 87% of them. Document precision 0.6162 → 0.8806.
 - **Still fast: 3.93 ms p95 on one core** against a 5 ms budget, reading 12,000 characters — 12× deeper than the prior champion's 1,000, and the depth is what makes it work on real documents.
 - **Nothing shipped.** Of five hard gates the best arm clears one. Document recall on real business documents (0.66–0.69) and 19 of 55 priority-tag recall pairs fall short. The gates were written before the numbers arrived and were not moved afterwards.
 - **The route is** budget_exhausted, **not** target_infeasible. Both failure clusters have named, unexplored mechanisms — see *What to do next*.
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Model scoreboard

Model	Dataset	F1@5	F1@3	F1@1	Docs/sec	ms/doc
quiet-cascade	10360_betterdataai_ner_silver_eval_10.36k					
quiet-cascade	10626_ai4privacy_pii_masking_eval_10.63k					
quiet-cascade	20000_pii_holdout_20.00k					
quiet-cascade	30000_pii2_eval_25.15k					
quiet-cascade	38937_openpii_pii_eval_38.94k					
quiet-cascade	4000_datax-dualjudge-evalset-1.32k					
quiet-cascade	5617_nemotron_eval_5.36k					
quiet-cascade	6589_govdocs2-dualjudge-eval20-3.53k					
quiet-nogate	10360_betterdataai_ner_silver_eval_10.36k					
quiet-nogate	10626_ai4privacy_pii_masking_eval_10.63k					
quiet-nogate	20000_pii_holdout_20.00k					
quiet-nogate	30000_pii2_eval_25.15k					
quiet-nogate	38937_openpii_pii_eval_38.94k					
quiet-nogate	4000_datax-dualjudge-evalset-1.32k					
quiet-nogate	5617_nemotron_eval_5.36k					
quiet-nogate	6589_govdocs2-dualjudge-eval20-3.53k					
champion_1k	10360_betterdataai_ner_silver_eval_10.36k					
champion_1k	10626_ai4privacy_pii_masking_eval_10.63k					
champion_1k	20000_pii_holdout_20.00k					
champion_1k	30000_pii2_eval_25.15k					
champion_1k	38937_openpii_pii_eval_38.94k					
champion_1k	4000_datax-dualjudge-evalset-1.32k					
champion_1k	5617_nemotron_eval_5.36k					
champion_1k	6589_govdocs2-dualjudge-eval20-3.53k					

Measured 2026-08-25.